

20260723.md: Check-in with Charleston

Project status

gLike

- Conda environment set up

- **IMPORTANT**

- Replace

```
from setuptools import dist, setup, Extension
```

```
# bootstrap numpy; can we workaroud this?
```

```
dist.Distribution().fetch_build_eggs(["numpy>=1.14.5"])
```

```
# should be fine now
```

```
import numpy as np
```

- With:

```
from setuptools import setup, Extension
```

```
import numpy as np
```

- `fetch_build_eggs` is deprecated

- Installation works.

```
(glike) bash-4.4$ python
Python 3.8.20 | packaged by conda-forge | (default, Sep 30 2024, 17:52:49)
[GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import glike
>>> exit()
```

- Progress-wise: I have run through the [tutorial](#) and am starting to look into the code.
 - Specifically of interest is the `scipy.optimize.minimize` -like `glike.maximize` functionality. I dug around a lot in this particular `scipy.optimize` function during my PhD
 - Off of the top of my head I can try implementing a few optimization strategies (to prioritize computational efficiency, global minima/maxima id, specific conditions):

- Constrained optimization in which we ensure that parameter constraints are satisfied, e.g., one parameter larger or smaller than another, or that parameters sum to a specific value
- Brute-force grid-based optimization, which should be better at finding global maxima with a trade off of efficiency
- L-BFGS-B method (limited memory bfgs), which will work better when starting with parameters close to the true (global) optimum. Different starting positions are known to "burrow" to a singular optimum with this strategy. Worse when the initial guess is worse.
 - Basic idea is optimize based on matrix of second derivatives but only store like 5 matrices in memory
- Direct search / Nelder-Mead (first-derivative) approach, which will work when far from any minimum; yielding a more robust search (for finding global minima), but much slower *search* and moderately faster *computation* due to lack of gradient information.
- Could even try a combination of something like Fmin (direct search) and L-BFGS-B, trading off speed and efficiency interchangeably to get over "dips" in the likelihood surface while gliding over the "easy to search" parts

Notes for Jon

- Command-line for login (must be on VPN): `ssh mahj@10.72.9.14` (IP for discovery2 CARC server works for me)
- Command-line for interactive node:


```
salloc --time=2:00:00 --cpus-per-task=8 --mem=16G --account=mahj_1946
```
- Computing workspace: `/home1/mahj`
 - **ASK CHARLESTON ABOUT THIS**
 - Myquota indicates that I have 100GB in this workspace, as well as a scratch with 10TB, and access to `project2/chia657_28` (almost full), `project2/haiman_625`, and `project2/minjuic_62`
 - `bash-4.4$ myquota`

Directory	Size	Used	Limit	Files	Used	Limit
<code>/home1/mahj</code>	227 KiB		100 GiB	7		1,910,000
<code>/scratch1/mahj</code>	0 Byte		10.00 TiB	0		2,000,000
<code>/scratch2/mahj</code>	missing directory					
<code>/project2/chia657_28</code>	39.99 TiB		40.00 TiB	3,340,830		240,000,000
<code>/project2/haiman_625</code>	302.93 TiB		320.00 TiB	9,718,268		1,920,000,000
<code>/project2/minhuic_62</code>	9.51 TiB		15.00 TiB	59,196		90,000,000
- `.bashrc` can be edited from home workspace.
 - `module purge`
 - `module load conda`

- `conda init bash`
- `source ~/.bashrc`
- It doesn't like when I try and edit from `nano`, which is annoying (segmentation fault)
- Full installation sequence:
 - `conda create -n glike python numpy=1.17.5 scipy pandas matplotlib tskit tsinfer msprime tqdm pip`
 - `conda activate glike`
 - `git clone https://github.com/Ephraim-usc/glike.git`
 - `cd glike`
 - **Make Changes to `setup.py`**
 - `python -m pip install --no-build-isolation -e .`

Native Hawaiian Data

- I can confirm access to the data.
- ```
(glike) bash-4.4$ pwd
/project2/chia657_28/datasets/private/MEC/WGS/CCDG_WGS
(glike) bash-4.4$ ls
annotation_of_functional_variants pass_subset_10721_geno02_dropSingletons pass_subset_11001_ind readme
```
- I have recieved detailed for SFTP transfer to the Michigan Imputation panel
  - User name: `sftp-imp-refpanels-xfer`
  - Password: `8gjF-jZxA-A26L-wkYE`
  - Host: `share.sph.umich.edu`
  - Port: `2222`
- **Check-in with Charleston**
  - I can go ahead and start transferring this via `lftp` and `sftp put`
    - I'm pretty sure there's two dedicated transfer nodes on CARC for this. Looking into it

## Standards for joint multi-sample variant calling

- I have recieved instructions from Xinran regarding how to start up a Terra account, but haven't yet done so.